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# The Mean Value Inequality

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June 12, 2026

**Definition 1.** PolygonalLine Let  $(E, \|\cdot\|)$  be a normed vector space over  $\mathbb{R}$ , with subset  $A \subseteq E$ . A *polygonal line* in  $A$  is a sequence of  $n + 2$  points  $u_0, \dots, u_{n+1}$  in  $E$ , such that each line segment connecting neighbouring vertices lies completely in  $A$ , i.e.

$$t \cdot u_i + (1 - t) \cdot u_{i+1} \in A \quad \forall i = 0, \dots, n \text{ and } t \in [0, 1].$$

**Lemma 2.** *PolygonalLine.trans def:polygonalLine* Let  $U$  be a subset of a normed vector space. Given a polygonal line in  $U$ , joining  $x \in U$  and  $y \in U$ , together with another polygonal line joining  $y$  and  $z \in U$ , then by connecting the two polygonal lines head-to-tail, the result is a polygonal line in  $U$ , joining  $x$  and  $z$ .

*Proof.* The construction of the composition of polygonal lines follows common sense: For the polygonal lines  $a$ , defined by  $(x, a_1, \dots, a_n, y)$ , and  $b$ , defined by  $(y, b_1, \dots, b_m, z)$ , we define the composition  $a \circ b$  by the sequence  $(x, a_1, \dots, a_n, y, b_1, \dots, b_m, z)$ .

It remains to show that each segment is still contained in  $U$ . This is trivial as the first  $n + 2$  vertices belong to  $a$  and are all contained in  $U$ . Similarly, the last  $m + 2$  vertices belong to  $b$ , and are all contained in  $U$  as well.  $\square$

**Theorem 3.** *polygonal\_connected\_of\_connectedlem : composition\_of\_polygonal\_line\_in\_U\_is\_an\_open\_connected\_set\_of\_a\_normed\_vector\_space*

*Proof.* The case where  $U = \emptyset$  is trivial. So we assume  $U$  to be non-empty. Fix a point  $x \in U$ , and consider the set

$$V := \{u \in U \mid \exists \text{ a polygonal line connecting } x \text{ and } u.\}$$

We want to show that  $U = V$ .

To show that, we prove that  $V$  is both closed and open in  $U$  and  $V$  is non-empty. The latter is true, as  $x$  must be in  $V$ .

For closedness, we show that  $V = \bar{V}$ , the closure of  $V$ . Fix a point  $a \in \bar{V}$ , then  $\exists \epsilon > 0$ , such that  $B_\epsilon(a) \cap V \neq \emptyset$ . Take a point  $b \in B_\epsilon(a) \cap V$ . Since  $b \in V$ , there is a polygonal line joining  $x$  and  $b$ . Since  $b \in B_\epsilon(a)$ , there is a line segment joining  $b$  and  $a$ , which is a polygonal line as well. By composing the two polygonal lines, we create a polygonal line joining  $x$  and  $a$ . Hence  $a \in V$ .

For openness, fix a point  $a \in V$ . We take an open ball centred at  $a$  and contained in  $U$ , as  $U$  is open. It can be shown that this open ball is contained in  $V$  as well. It suffices to show that every point  $b$  in the open ball can be connected to  $x$  using a polygonal line, which can be constructed by composing the polygonal line joining  $x$  and  $a$  with the line segment joining  $a$  and  $b$ .

Since  $V$  is both open and closed in  $U$ , it is either equal to  $U$  or is the empty set. Since it is non-empty, we must have  $U = V$ .  $\square$

**Definition 4.** *thm:existence\_of\_polygonal\_line* Let  $U$  be an open connected set of a Banach space  $E$ . For  $a, b \in U$ , we define  $d_U(a, b)$  to be the infimum of the lengths of the polygonal lines contained in  $U$  joining  $a$  and  $b$ .

**Theorem 5.** *def:distance in U lem:composition\_of\_polygonal\_line\_thm : polygonal\_line\_length\_at\_least\_interval\_length is a distance in the topological space U.*

*Proof.* Let  $\varphi$  be an arbitrary polygonal line joining  $a$  to  $b$  in  $U$ , defined by the sequence  $a_0, \dots, a_{n+1}$ , with  $a_0 = a$  and  $a_{n+1} = b$

*Positivity:* The length of  $\varphi$  is  $\text{Length}(\varphi) := \sum_{i=0}^n \|a_{i+1} - a_i\|$ , which by the definition of a norm function is strictly non-negative. Since  $d_U(a, b)$  is the infimum of a strictly non-negative set, it must itself be non-negative. If  $d_U(a, b) = 0$ , by the previous theorem we have  $\|b - a\| \leq d_U(a, b) = 0$ , which implies  $a = b$ .

*Symmetry:* For any polygonal line  $\varphi$  from  $a$  to  $b$  in  $U$ , the reverse line  $\psi$  defined by  $a_{n+1}, \dots, a_0$  is a polygonal line from  $b$  to  $a$ .  $\text{Length}(\psi) = \sum_{i=0}^n \|a_i - a_{i+1}\| = \sum_{i=0}^n \|a_{i+1} - a_i\| = \text{Length}(\varphi)$ , so the set of lengths of polygonal lines from  $a$  to  $b$  and from  $b$  to  $a$  are equivalent and therefore  $d_U(a, b) = d_U(b, a)$ .

*Triangle Inequality:* Let  $\varphi$  be a polygonal line from  $a$  to  $b$  and  $\psi$  be a polygonal line from  $b$  to  $c$ . The composition of  $\varphi$  and  $\psi$  is a polygonal line  $\theta$  from  $a$  to  $c$ , where

$$\begin{cases} \varphi \text{ is defined by } (a, a_1, \dots, a_n, b) \\ \psi \text{ is defined by } (b, b_1, \dots, b_m, c) \\ \theta \text{ is defined by } (a, a_1, \dots, a_n, b, b_1, \dots, b_m, c) \end{cases}$$

By the definition of infimum,  $d_U(a, c) = \inf \text{Length}(\theta) \leq \text{Length}(\theta) = \text{Length}(\varphi) + \text{Length}(\psi)$ . Since  $\varphi$  and  $\psi$  are arbitrary, we take the infimum on the right hand side, and hence  $d_U(a, c) \leq d_U(a, b) + d_U(b, c)$ .  $\square$

**Theorem 6.** *def:distance in U thm:existence\_of\_polygonal\_line\_norm\_image\_sub\_epsilon\_norm\_has\_F\_Deriv\_Within\_epsilon' Let  $f : U \rightarrow F$  be a differentiable mapping with values in a Banach space  $F$ . Assume that*

$$\|f'(x)\| \leq k \quad \forall x \in U$$

*then for any  $x_1, x_2 \in U$ ,  $\|f(x_1) - f(x_2)\| \leq k \cdot d_U(x_1, x_2)$ .*

*Proof.* Let  $x_1, x_2 \in U$  and  $l$  a polygonal line contained in  $U$  joining  $x_1$  and  $x_2$ . Suppose  $l$  is defined by  $n + 1$  vertices  $a_0, \dots, a_{n+1}$ , where  $a_0 = x_1, a_{n+1} = x_2$  and  $a_i \in U$ . Then from the mean value inequality on each interval  $(a_i, a_{i+1})$ , we have  $\|f(a_i) - f(a_{i+1})\| \leq k \cdot \|a_i - a_{i+1}\|$ . From the triangle inequality we have

$$\begin{aligned} \|f(x_1) - f(x_2)\| &\leq \sum_{i=0}^n \|f(a_i) - f(a_{i+1})\| \\ &\leq \sum_{i=0}^n k \cdot \|a_i - a_{i+1}\| \\ &= k \cdot \text{Length}(l) \end{aligned}$$

as  $\text{Length}(l)$  is the sum of the length of each interval. Therefore, taking the infimum we get

$$\|f(x_1) - f(x_2)\| \leq k \cdot \inf \text{Length}(l) = k \cdot d_U(x_1, x_2)$$

□

**Definition 7.** Let  $(E, \|\cdot\|)$  be a normed vector space over  $\mathbb{R}$ . The *length of an interval* with origin  $a \in E$  and end  $b \in E$  is  $d(a, b) := \|b - a\|$ .

**Definition 8.** `def:polygonalLine def:intervalLength` Given a polygonal line  $\varphi$  defined by  $(a_0, \dots, a_{n+1})$ , with some fixed  $n \geq 0$  the *length of the polygonal line*  $\varphi$  is the sum of the lengths of the intervals from which it is formed:

$$\text{Length}(\varphi) := \sum_{i=0}^n d(a_i, a_{i+1}) = \sum_{i=0}^n \|a_{i+1} - a_i\|$$

**Lemma 9.** *Given a sequence  $(a_i)_{i=0}^n$  with  $a_i \in E$  for some  $n \geq 1$ ,  $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$*

*Proof.* • **Base Case ( $n = 1$ ):** For  $n = 1$ , we need  $\|a_1 - a_0\| \geq \|a_1 - a_0\|$  which follows immediately by reflexivity.

- **Induction Hypothesis:** Assume that the result holds for any such sequence  $(b_i)_{i=0}^k$ . That is,

$$\sum_{i=0}^{k-1} \|b_{i+1} - b_i\| \geq \|b_k - b_0\|$$

- **Induction Step:** We want to show that  $\sum_{i=0}^k \|a_{i+1} - a_i\| \geq \|a_{k+1} - a_0\|$  follows from this assumption. For some arbitrary sequence  $(a_i)_{i=0}^{k+1}$ ,

$$\begin{aligned} \|a_{k+1} - a_0\| &= \|(a_{k+1} - a_k) + (a_k - a_0)\| \\ &\leq \|a_{k+1} - a_k\| + \|a_k - a_0\| \\ &\leq \|a_{k+1} - a_k\| + \sum_{i=0}^{k-1} \|a_{i+1} - a_i\| \\ &= \sum_{i=0}^k \|a_{i+1} - a_i\| \end{aligned}$$

Therefore, by the principle of mathematical induction,  $\sum_{i=0}^{n-1} \|a_{i+1} - a_i\| \geq \|a_n - a_0\|$  for any such sequence with any  $n \geq 1$ . □

**Theorem 10.** *lemma: norm<sub>s</sub>um<sub>a</sub>tleast<sub>s</sub>um<sub>n</sub>ormThe length of a polygonal line  $\varphi$  defined by  $a_0, \dots, a_{n+1}$  for some fixed  $n \geq 0$  between origin  $a = a_0$  and end  $b = a_{n+1}$  is at least equal to the length of the interval between them:*

$$\sum_{i=0}^n \|a_{i+1} - a_i\| \geq \|b - a\|$$

*Proof.* The result follows immediately from the lemma.

□